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The Concept of the Photon in Question: The Controversy Surrounding the HBT Effect circa 1956–1958

ABSTRACT

The history of the concept of the photon in twentieth-century physics is far from a simple story opening with Einstein's vision of light as a collection of indivisible particles whose energy and momentum are conserved during its interaction with matter, and reaching closure with the wave-particle duality as an accomplishment of quantum mechanics. Since then there has been an intermittent debate on the need for and adequacy of such a concept, even if this debate has been absent from the literature on the history of physics and from physics teaching. This paper analyzes a major event which led to the revival of this debate, namely, the experiment carried out by Robert Hanbury Brown and Richard Quentin Twiss (HBT) in 1956 in the context of low-intensity interferometry. As part of their work to build a new kind of interferometer to measure the diameter of optical stars, their results seemed to suggest that photons split through two different channels and detectors. These results stirred up a debate involving Edward Purcell, Eric Brannen, Harry Ferguson, Peter Fellgett, Richard Sillitto, Lajos Jánossy, Leonard Mandel, and Emil Wolf, in addition to Hanbury Brown and Twiss themselves. The building of this device in astronomy thus renewed the old controversy about the nature of light. Later on, with the invention of lasers, the HBT experimental results played a role in developments leading to the creation of quantum optics and currently play a role in various fields in physics.

KEY WORDS: Eric Brannen, Robert Hanbury Brown, Edward M. Purcell, photon concept, wave-particle duality, nature of light, interferometry

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INTRODUCTION

As is well known, after the early development of quantum theory, radiation was defined as a collection of indivisible particles—photons—whose energy and momentum were conserved during its interaction with matter.[1] However, an astonishing and unexpected experimental result, published thirty years after the creation of the quantum theory, called the canonical concept of the photon into question. This experiment was carried out in 1956 by the British scientists Robert Hanbury Brown (1916–2002) and Richard Quentin Twiss (1920–2005). Hanbury Brown and Twiss were neither part of the community involved in discussions about the foundations of quantum theory nor researchers investigating the fundamental concepts of physics. Rather, they were involved in applying physical concepts to astronomy and consequently introducing new methods in that field. Their work "put the cat among the pigeons."[2] Indeed, HBT's results stirred up a heated controversy in the community of physicists.[3] Hanbury Brown and Twiss were awarded the Albert Michelson Medal in 1982 for their work of 1956.

In the Hanbury Brown–Twiss (HBT) experiment, a low-intensity beam of light was split into two components by a half-silvered mirror, and then the

1. For instance, see: Olivier Darrigol, "A Simplified Genesis of Quantum Mechanics," *Studies in History and Philosophy of Modern Physics* 40, no. 2 (2009): 151–66; Stephen Brush, "How Ideas Became Knowledge: The Light-Quantum Hypothesis 1905–1935," *HSPS* 37, no. 2 (2007): 205–46; Helge Kragh, *Quantum Generations: A History of Physics in the Twentieth Century* (Princeton, NJ: Princeton University Press, 1999); Max Jammer, *The Conceptual Development of Quantum Mechanics* (New York: McGraw-Hill, 1966); Bruce R. Wheaton, *The Tiger and the Shark: Empirical Roots of Wave-Particle Dualism* (Cambridge: Cambridge University Press, 1983); Roger H. Stuewer, *The Compton Effect: Turning Point in Physics* (New York: Science History Publications, 1975); Thomas S. Kuhn, *Black-Body Theory and the Quantum Discontinuity, 1894–1912* (Chicago: University of Chicago Press, 1978).

2. Robert Hanbury Brown, *Boffin: A Personal Story of the Early Days of Radar: Radioastronomy and Quantum Optics* (New York: Taylor & Francis Group, 1991), 117–34, on 120.

3. Other scholars have also highlighted the controversial debate about the Hanbury Brown and Twiss experiment: David Owen Edge and Michael Joseph Mulkay, *Astronomy Transformed: The Emergence of Radioastronomy in Britain* (New York: Wiley-Interscience, 1976), on 146; B. Lovell and R. M. May, "Robert Hanbury Brown (1916–2002)," *Nature* 416 (2002): 34; J. Davis and B. Lovell, "Robert Hanbury Brown 1916–2002," *Historical Records of Australian Science* 14 (2003): 459–83, on 469; B. Tango, "Richard Quentin Twiss 1920–2005," *Astronomy & Geophysics* 47 (2006): 4.38; Mario Bertolotti, *Masers and Lasers—An Historical Approach* (Bristol: Adam Hilger, 1983), on 203; Joan Bromberg, "Modelling the Hanbury Brown Twiss Effect—The Mid-Twentieth Century Revolution in Optics" (paper presented at the HQ–3 Conference on the History of Quantum Physics, Berlin, June 28–July 2, 2010).

components were detected separately through two photomultipliers. Hanbury Brown and Twiss claimed that two photons had been detected at the same time. Because of the intensity of the source used, it had been expected that only single photons were arriving at the mirror in a certain time interval. Thus, a question arose: how could they find a correlation between photons if, as proposed by Einstein and spread widely by quantum theory textbooks, these are indivisible particles? From the perspective of the conventional concept of the photon, the HBT experimental result seemed to go against the foundations of the quantum theory because there would be no way to detect two photons at the same time at two different detectors, assuming a photon as a small localized indivisible particle. The debate over the HBT results was intense, so intense, indeed, that, as remarked later by Hanbury Brown, some physicists even went as far as to claim that he and Twiss misunderstood the quantum theory.[4]

In hindsight, physicists can easily see what was at stake. Nowadays, physics makes a clear distinction between attenuated light and light described by number states (Fock states), that is, states with a well-defined number of photons. For the former, the probability of finding n photons is given by a Poisson distribution, and there is indeed a slim chance of getting two photons at a time. As for the latter, in the case of a single-photon state where such a distribution is not valid, the probability of finding n photons (with $n \neq 1$) is null. The HBT sources were of the former kind. However, physicists came to understand this after applying quantum field methods to optics and developing quantum optics. Surely, it would be an anachronism to use a current explanation for what happened at that time.[5] Our argument is that, in fact, the HBT experimental results questioned the standard concept of the photon proposed by the old quantum theory, arousing a heated experimental and theoretical controversy.

4. Robert Hanbury Brown, "Paraboloids, Galaxies and Stars: Memories of Jodrell Bank," in *The Early Years of Radioastronomy: Reflections Fifty Years after Jansky's Discovery*, ed. W. T. Sullivan III (Cambridge: Cambridge University Press, 1984), 213–35, on 230. While it is more precise to refer to the social phenomenon under study as a scientific controversy, from time to time we describe it as a debate or dispute, each of which has a looser meaning. On scientific controversies, see H. Tristram Engelhardt, Jr. and Arthur L. Caplan, eds., *Scientific Controversies: Case Studies in the Resolution and Closure of Disputes in Science and Technology* (Cambridge: Cambridge University Press, 1987), and the special issue on "Controversies" in *Science in Context* 11, no. 2 (1998).

5. R. G. W. Brown and E. R. Pike, "A History of Optical and Optoelectronic Physics in the Twentieth Century," in *Twentieth Century Physics*, vol. III, ed. Laurie Brown, Abraham Pais, and Sir Brian Pippard (New York: American Institute of Physics and Bristol/Philadelphia: Institute of Physics Publishing, 1995), on 1421–22, 1439–41, and 1459–60.

Although the quantum statistics of radiation had been developed by Satyendra Nath Bose (1894–1974) and by Albert Einstein (1879–1955), resulting in "the abandonment of the classical concept of individually identifiable particles," in the late 1950s physicists still interpreted photons as distinguishable entities—indivisible objects—as we will see during the HBT debate.[6] At that time, three concepts of the photon emerged: photons as indivisible entities, photons as bosons constituting the electromagnetic field, and photons as wave packets. Even though our focus is on how the HBT experiment encouraged physicists to revisit the concept of the photon, it is important to mention that the HBT results also played a fundamental role in the renaissance of a classical discipline—optics—which had not been developing so fully for a long time.[7] After the HBT experiment and the development of the laser, physicists turned their attention to optics again, contributing experimentally and theoretically to its renaissance.

In addition to contributing to the development of quantum optics, astronomy, and optics at the time, the HBT effect is currently part of vanguard contemporary physics in different fields such as high-energy physics, nuclear physics, atomic physics, and condensed matter physics. As the Web of Science indicates, for instance, Hanbury Brown and Twiss's 1956 paper has 509 citations, of which 259 are in articles published in the twenty-first century. Even in the early 1960s, physicists envisioned applications of the HBT effect into military devices. In fact, Marvin Goldberger, Kenneth Watson, and Hal Lewis, working for the American defense with the Jason project, used this effect with radar in order to measure the size of incoming warheads.[8]

In this paper, we examine the 1956 Hanbury Brown and Twiss episode, their experimental results and theoretical model, the repercussions in the physics community, and the controversy surrounding their work between 1956 and 1958. In particular, we investigate the conceptual debate, especially about the nature of the photon, which the HBT experiment stirred up. In the first section, we present the background to Hanbury Brown and Twiss and their experiment and results; the next section focuses on the controversy surrounding this experiment circa 1956–58; the third section is dedicated to the responses given by our protagonists to criticism; and finally, we briefly consider the influence of the HBT experiment on the early development of quantum optics.

6. Olivier Darrigol, "Statistics and Combinatorics in Early Quantum Theory, II: Early Symptoma of Indistinguishability and Holism," *HSPS* 21, no. 2 (1991): 237–98, on 239.

7. Kragh, *Quantum Generations* (ref. 1), 390.

8. This research was carried out on 20 Aug 2012. See also Ann Finkbeiner, *The Jasons – The Secret History of Science's Postwar* Elite (New York: Penguin, 2006), 51.

THE HBT EXPERIMENT

The HBT experiment appeared in the context of astronomical interferometry, a field which was born in the nineteenth century following the acceptance of the wave nature of light. Interferometers exploit the nature of waves to superimpose waves and thus obtain information about them. In this case, the formation of interference fringes occurs when superimposed waves have phases whose differences are constant. Indeed, interferometry became a widely used technique in science and engineering and led the 1907 Nobel Prize to be awarded to Albert A. Michelson (1852–1931) precisely "for his optical precision instruments and the spectroscopic and metrological investigations carried out with their aid."[9]

In addition to these applications, and following an early suggestion of Armand H. Fizeau (1819–96), Michelson launched the use of interferometers for measuring the angular diameters of stars. By 1890, Michelson had already developed the theory for this use. He showed that by masking the objective of a telescope except for two slits separated by a distance b, the fringes would disappear when $\alpha = \lambda/b$, where α is the angular diameter and λ the wavelength of the incoming light. As this magnitude is also the resolution of a telescope with aperture b, and there are limitations to building telescopes with larger apertures, Michelson thought of using two mirrors of a refractometer to capture light and then send the two beams through a system of mirrors to be recorded. The two signals are then multiplied, giving a product with regular maxima and minima that are the equivalent to the fringes in the traditional Michelson interferometer. The fringe visibility depends on the separation of the mirrors and the angular diameter of the source, and it may be obtained through the Fourier transform of the brightness distribution of the source. The distance between mirrors could be greater than the size of the telescope objectives, thus amplifying the resolution. With such a device Michelson hoped to circumvent the effects of atmospheric turbulence that had so far limited the resolution of optical telescopes. The Michelson stellar interferometer, however, spent thirty years in hibernation. In 1920, after a time delay which has intrigued historians, the apparatus of the Mount Wilson Observatory was eventually used by Michelson and F. Pease to measure the diameter of the red giant star Betelgeuse.[10]

9. Nobelprize.org, "The Nobel Prize in Physics 1907," at http://www.nobelprize.org/nobel_prizes/physics/laureates/1907/ (accessed 11 Dec 2012).

10. M. H. Cohen et al., "Radio Interferometry at One-Thousandth Second of Arc," *Science* 162, no. 3849 (1968): 88–94; Albert A. Michelson, "On the Application of Interference Methods to Astronomical Measurements," *Philosophical Magazine* 30, no. 182 (1890): 1–21; David H.

It was in the early 1930s that Karl Jansky (1905–50), working as an engineer at Bell Laboratories, recorded the first reception of radio waves coming from outer space. After the war, radio astronomy rapidly became a subfield of astronomy, with scientists drawing most of their skills and machinery from wartime experience with radar. It comes as no surprise, then, as its chroniclers have put it, that, according to Fujinobu Takahashi, "the initial advances in radio astronomy were achieved by the astronomers of the victors in World War II." Michelson's stellar interferometer was then converted to measure angular diameters of radio cosmic sources, but this new application of interferometry came at a cost, as we will see.[11]

Both Hanbury Brown and Twiss were born in India, which was then part of the British Empire. In 1935, Hanbury Brown received his B.Sc. in electrical engineering from the University of London and spent eleven years working on the secret development of radar for Britain's Air Ministry. During World War II, he also conducted research on radar at the U.S. Naval Research Laboratory in Washington, DC. During 1947–49, Hanbury Brown became a consulting engineer in the field of radar for companies in France, the United States, and the United Kingdom, having as a senior partner Robert Watson-Watt (1892–1973), a Scottish physicist who had played an important role in the wartime development of radar in the U.K. In 1949, Hanbury Brown began research for a doctoral degree at the Jodrell Bank radioastronomy research center at the University of Manchester.[12]

Twiss, who was slightly younger, completed the Mathematical Tripos at Cambridge in 1941 and received his doctoral degree from MIT in 1949, working on the theory of magnetrons. During WWII, Twiss worked on radar in the Admiralty, the Naval Service of the British Armed Forces. Upon returning to

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DeVorkin, "Michelson and the Problem of Stellar Diameters," *Journal for the History of Astronomy* 6 (1975): 1–18.

11. Edge and Mulkay, *Astronomy Transformed* (ref. 3); Fujinobu Takahashi et al., *Very Long Baseline Interferometer* (Tokyo: Ohmsha, 2000), on 1.

12. J. Davis and B. Lovell, "Robert Hanbury Brown 1916–2002," *Historical Records of Australian Science* 14 (2003): 459–83, on 462–63 and 464; B. Lovell and R. M. May, "Robert Hanbury Brown (1916–2002)," *Nature* 416 (2002): 34. For details about the radar in World War II, see H. E. Guerlac, *Radar in World War II* (New York: American Institute of Physics, 1987). For studies on the Jodrell Bank, consult B. Lovell, *The Story of Jodrell Bank* (Oxford: Oxford University Press, 1968); B. Lovell, *Out of the Zenith: Jodrell Bank, 1957–70* (Oxford: Oxford University Press, 1973); B. Lovell, *The Jodrell Bank Telescopes* (Oxford: Oxford University Press, 1985); J. Agar, *Science & Spectacle: The Work of Jodrell Bank in Post-War British Culture* (Amsterdam: Harwood Academic, 1998).

the United Kingdom after getting his degree from MIT, Twiss carried out research on electromagnetic radiation and in 1955 was part of the research group at the Division of Radiophysics in Sydney, Australia.[13]

It was at Jodrell Bank that Hanbury Brown started to do research on radio astronomy. In the early 1950s, the institute had changed its research focus, moving from meteor studies to research on radio stars, which contributed significantly to the development of radio astronomy.[14] At that time, one of the problems faced by radio astronomers was to determine the ontology of the sky. This could be done by measuring the angular diameter of objects, and, depending on the value found, they could be characterized as nebulae, galaxies, or stars.[15]

The Michelson interferometer, indeed, a Michelson stellar interferometer adapted for radio sources, was used to measure the angular diameter of the objects at the end of the 1940s. In this apparatus (Fig. 1) the two vertical aerials *A* and *B* detected the radio signals which transited the plane normal to a horizontal baseline *L* (a conventional cable). The outputs were detected by a receiver connected to the center of the cable, and then registered by a power recorder; the oscillations observed during the transit of the source were similar to the interference fringes in a Michelson stellar interferometer.[16] In the case of a large angular size, it would be necessary to have a small separation between the aerials *A* and *B* to measure the angular diameter. For small ones, however, the separation of the aerials would have to be extremely large.

There were two major problems with this kind of interferometer. The horizontal baseline between the two aerials could be extended to fifty kilometers without causing any perturbation in the system. Nevertheless, due to the instability of the phase in the transmission process through the cables, the measurements of the angular diameter might be especially inaccurate when using long baselines.[17] This limitation on the size of the separation restricted the use of the Michelson interferometer for only some radio sources. Moreover, using a very long cable to connect the aerials made the interferometer "both cumbersome and expensive." The other difficulty with the Michelson

13. Tango, "Richard Quentin Twiss" (ref. 3), 4.38.

14. Edge and Mulkay, *Astronomy Transformed* (ref. 3), 19–20.

15. R. Hanbury Brown, interview by Ragbir Bhathal, 1995, RHB, Box 2, Section A.31, on 11–12.

16. Robert Hanbury Brown and Richard Q. Twiss, "A New Type of Interferometer for Use in Radioastronomy," *Philosophical Magazine* 45 (1954): 663–82, on 664.

17. Robert Hanbury Brown, R. C. Jennison, and M. K. Das Gupta, "Apparent Angular Sizes of Discrete Radio Sources," *Nature* 170 (1952): 1061–63, on 1061.

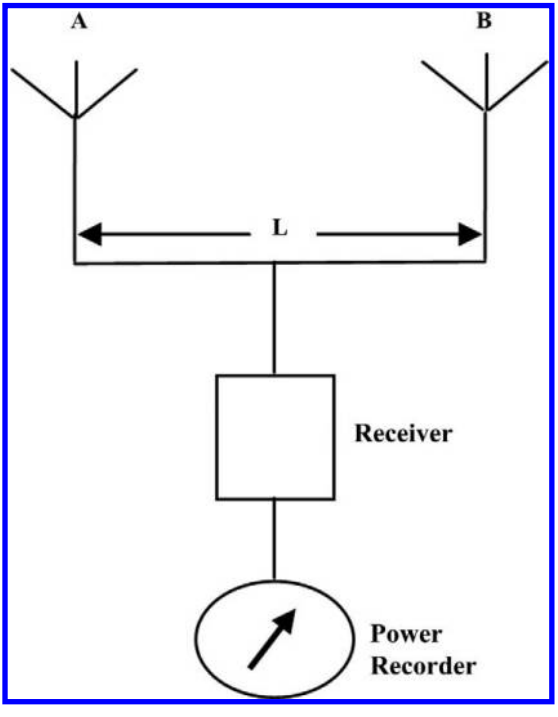


FIG. 1: Simplified diagram of Michelson's interferometer used in radio astronomy. *Source*: Hanbury Brown and Twiss, "New Type of Interferometer" (ref. 16), 664.

interferometer was associated with its excessive sensitivity to ionospheric effects that might interfere with the measurements.[18]

The construction of a new type of interferometer able to work with a long baseline required significant engineering skills and new techniques. Hanbury Brown demonstrated these in constructing the "intensity interferometer," the use of which would make it possible to compare the intensities at two different points of an electromagnetic field, instead of comparing the amplitude and phase of the oscillations in the Michelson interferometer. As he later recounted,

> [L]ate one night in 1949 I was wondering whether, if I were to take "snapshots" of the noise received from a radio source on oscilloscopes at the outputs

18. Hanbury Brown and Twiss, "New Type of Interferometer" (ref. 16), 663–64, 667, 678. The ionosphere, composed of free electrons, is able to cause the reflection and absorption of radio wave changing, thus, the radiation intensity arriving at the earth's surface.

> of two spaced receivers, I could compare these snapshots. The answer to that question led me directly to the idea of an interferometer in which the *intensities* of two noise-like signals are compared instead of their amplitude and phase.[19]

The idea of an intensity interferometer was thus born. However, Hanbury Brown needed a sophisticated theoretical model for it. He therefore asked his friend Vivian Bowden to find someone who could help with the mathematics to model the intensity interferometer. According to Hanbury Brown, "[u]nfortunately I didn't know enough mathematics to work out the answer . . . Vivian found me someone called Richard Twiss who, like me, was born in India of an Army family, but, unlike me, was a talented mathematician."[20] The ensuring collaboration between Hanbury Brown and Twiss is a not unfamiliar example of interaction between an experimentalist and theoretician working together in order to construct a new instrument. This collaboration, however, was conducted at a distance: Hanbury Brown was part of the Jodrell Bank group at the University of Manchester, and Twiss worked for the Services Electronics Research Laboratory at Baldock located thirty-eight miles from London. Even though they had some face-to-face meetings, their collaborative work was conducted by postal correspondence.

During his first visit to the University of Manchester, Hanbury Brown explained to Twiss his idea for a new type of interferometer and asked him to verify mathematically how sensitive the new instrument might be. Afterwards, working on the mathematics, Twiss concluded that (as later remembered by Hanbury Brown), "[t]his idea of yours is no good, it doesn't work!" However, when Hanbury Brown and Twiss were reviewing Twiss's calculations, they found that there was a small mistake in one of the integrals, and after correcting it, there was no doubt that the interferometer would work properly. Nonetheless, it would not be sufficiently sensitive to obtain the angular diameter of most radio sources, only the two strongest ones, Cygnus and Cassiopeia. While Twiss was working on the theory for the new interferometer in detail, Hanbury Brown, with the assistance of his research students Roger C. Jennison and Mrinal K. Das Gupta, constructed the new intensity interferometer.[21]

By 1950 the intensity interferometer had been built. In this instrument (Fig. 2) the signals were detected by two aerials A_1 and A_2, which were

19. Hanbury Brown, "Paraboloids, Galaxies and Stars" (ref. 4), 226–27.
20. Hanbury Brown, *Boffin* (ref. 2), 105.
21. Ibid., 105–06.

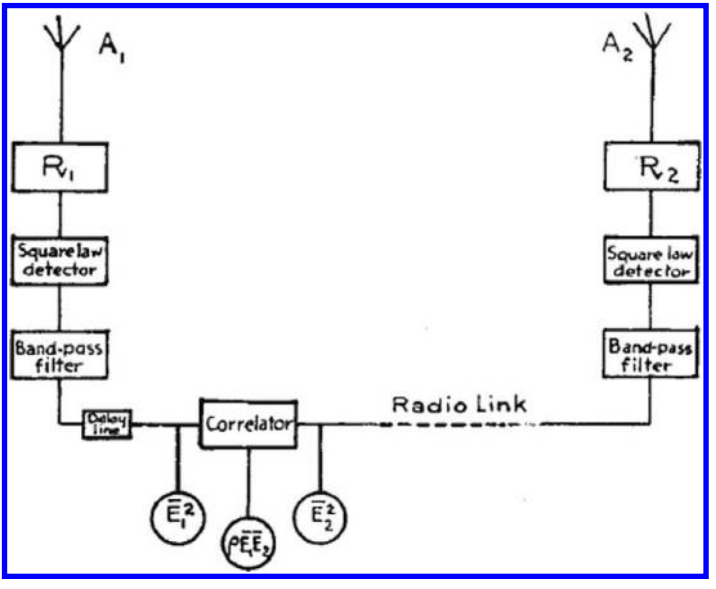


FIG. 2: Outline of the new type of interferometer developed by Hanbury Brown at Jodrell Bank. *Source*: Hanbury Brown, Jennison and Das Gupta, "Apparent Angular Sizes" (ref. 17), 1061.

connected through two independent receivers R_1 and R_2; the outputs were separately rectified in each square-law detector (output voltage proportional to the square of the electric field) and then passed through the two low-frequency filters; the two low-frequency outputs were multiplied together in a correlator, thus obtaining their "cross-correlation," which would be the measurement of the similarity of two different signals, or waveforms. Hanbury Brown and Twiss derived the following expression for the cross-correlation coefficient, which is analogous to that for the visibility of the fringes in the Michelson stellar interferometer:

$$\rho = \frac{sin^2\left(\frac{\pi\alpha b}{\lambda}\right)}{\left(\frac{\pi\alpha b}{\lambda^2}\right)} \qquad (1)$$

where α is the angular width of an equivalent rectangular source of constant surface intensity, b is the length of the baseline, and λ is the wavelength.[22] Equation (1) means that if a suitable baseline is chosen, it is possible to obtain a value for the angular diameter of the source.

22. Hanbury Brown, Jennison, and Das Gupta, "Apparent Angular Sizes" (ref. 17), 1062.

The correlator, comparing the two signals using the technique of Fourier analysis, was able to isolate specific components of a compound waveform. In order to use short baselines, a radio link could be inserted between the two aerials; for extremely long baselines the signals could be recorded on tapes and correlated later. This was an advantage compared to the Michelson interferometer, whose inaccuracy was exacerbated when using very long baselines. In their first theoretical model for the new interferometer, Hanbury Brown, Jennison, and Das Gupta calculated the cross-correlation as a function of the apparent angular diameter of the radio sources, the effective length of the baseline, and the wavelength.[23]

Even though the new interferometer successfully measured the angular diameter of Cassiopeia and Cygnus when compared to other measurements available, Hanbury Brown recounted later that he was deeply disappointed at the final result. Because it was not necessary to use a long baseline—only a few kilometers—to measure the angular diameter of these sources, "there was no need to have developed the intensity interferometer; we could have done the same job with a conventional interferometer in half time and with half effort. We had built a steam-roller to crack a nut."[24] The effort involved in the construction of an intensity interferometer thus seemed to have been unnecessary. Nonetheless, one interesting finding brought Hanbury Brown and Twiss to the realization that such an interferometer could be used to measure the angular diameter not of radio-emitting stars, but of bright, visible stars.

Observing the intensity interferometer in use during one of Twiss's visits to Jodrell Bank, Twiss and Hanbury Brown realized that it was working successfully, even though "on that particular day the signal was scintillating violently as it passed through the ionosphere and [they] noticed that, although the strength of the signals in the two antennae were fluctuating wildly, their correlation was unchanged."[25] Then, by analogy, they conjectured that the intensity interferometer seemed to be able to work accurately in a turbulent medium, that is, with the fluctuation density of the atmosphere. This had been a limitation of the Michelson stellar interferometer, when used for optical sources, in addition to the issue of long baselines. Because of that advantage, Hanbury Brown and Twiss decided to construct an interferometer for optical

23. Ibid., 1061; Hanbury Brown, "Paraboloids, Galaxies and Stars" (ref. 4), 227.

24. Hanbury Brown, "Paraboloids, Galaxies and Stars" (ref. 4), 228.

25. Ibid., 228; Hanbury Brown, *Boffin* (ref. 2), 118; Hanbury Brown, interviewed by Ragbir Bhathal (ref. 15), 117.

astronomy using the same principles as the intensity interferometer in radioastronomy, allowing measurements of the angular diameter of bright stars to be made.

By 1954, Hanbury Brown and Twiss shifted their interest from radioastronomy to the domain of optics. During the second half of the 1950s, as argued by the historians of science David Edge and Michael Mulkay, "radio astronomers began to extend the application of their techniques to areas that previously had been exclusively optical."[26] In the winter of 1956, Hanbury Brown and Twiss published an article entitled "The Correlation between Photons in Two Coherent Beams of Light," coherence implying that there is a constant phase relationship between two values of the electromagnetic field at separated points or separated times. In this paper, they reported a laboratory test to verify if the same techniques and principles used in the HBT intensity interferometer for radio astronomy could be applied to optical astronomy. Such a test was carried out with an artificial source of light, a high-pressure mercury arc. In doing so, Hanbury Brown and Twiss left in fact the field of radio astronomy and turned to optics.

In the HBT intensity interferometer, working with radio waves, a correlation between intensity fluctuations at two different points could be obtained. Nevertheless, acknowledging "this fundamental effect has never been directly observed with light, and indeed its very existence has been questioned," Hanbury Brown and Twiss decided to perform a laboratory test before building their stellar interferometer to investigate whether or not there would be a correlation between beams of light.[27]

In the HBT optical system (Fig. 3), the light source was focused by a lens and sent through a system of filters. The beam of light was divided by a semitransparent mirror to focus on the cathodes of the photomultipliers C_1 and C_2. The fluctuations in the output were amplified and multiplied together in a correlator. The correlation in the fluctuations was thus obtained through an integrating motor. The photomultiplier C_1 could move vertically and consequently the measurements could be obtained in two different ways: first, when the optical paths from the mirror to the cathodes C_1 and C_2 were at the same length, that is, superimposed when viewed from the source; and

26. Edge and Mulkay, *Astronomy Transformed* (ref. 3), 277.

27. Robert Hanbury Brown and Richard Quentin Twiss, "Correlation between Photons in Two Coherent Beams of Light," *Nature* 177, no. 4497 (1956a): 27–29, on 27; Hanbury Brown, "Paraboloids, Galaxies and Stars" (ref. 4), 229.

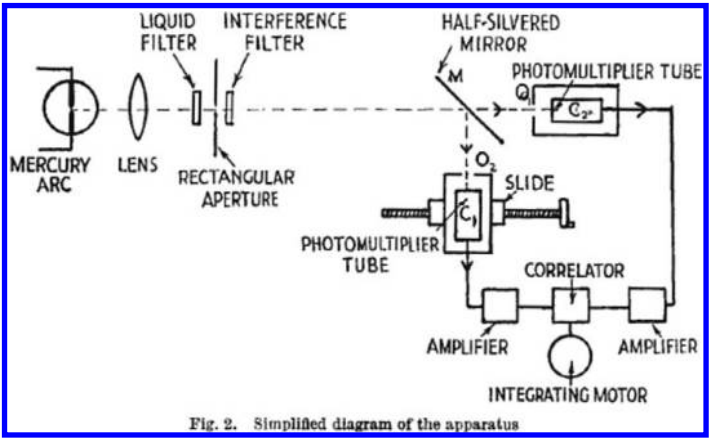


FIG. 3: The optical system set-up by Hanbury Brown and Twiss. *Source*: Hanbury Brown and Twiss, "Correlation between Photons" (ref. 27), 28.

second, when those paths were not the same length but rather separated by a distance d.[28]

The theoretical model for the HBT optical interferometer was calculated using a semiclassical approach. Hanbury Brown and Twiss considered radiation from the mercury source as a classical wave, but they also used the quantization of the radiation in the photoelectric emission for photodetection. Assuming the probability of the emission of a photoelectron to be proportional to the square of the amplitude of the incident light, Hanbury Brown and Twiss calculated the correlations between the fluctuations in the current from the cathodes by using classical electromagnetic wave theory. In their theoretical studies they obtained first the correlation $S(0)$ when the two cathodes were superimposed,[29]

$$S(0) = A.T.b_\nu.f\left(\frac{\alpha_1\theta_1\pi}{\lambda_0}\right).f\left(\frac{\alpha_2\theta_2\pi}{\lambda_0}\right)\int \alpha^2(\nu).n_0^2(\nu).d\nu, \qquad (2)$$

then determined the associated root-mean-square fluctuations N,

$$N = A.T.\frac{2m}{m-1}.b_\nu(b_\nu T)^{-\frac{1}{2}}\int \alpha(\nu).n_0(\nu).d\nu \qquad (3)$$

28. Hanbury Brown and Twiss, "Correlation between Photons" (ref. 27), 27–28.
29. Ibid., 28.

In these equations, A is a constant of proportionality, T is the time of observation, $\alpha(\nu)$ is the quantum efficiency of the photocathodes at a frequency ν, $n_0(\nu)$ is the number of quanta incident, b_ν is the bandwidth of the amplifiers, $m/(m-1)$ is the excess noise, a_1 and a_2 are the horizontal and vertical dimensions of the photocathode apertures, θ_1 and θ_2 are the angular dimensions of the source as viewed from the photocathodes, and λ_0 is the mean wavelength of the light. Hanbury Brown and Twiss mentioned that the factor

$$f\left(\frac{\alpha\theta\pi}{\lambda_0}\right)$$

could be found through the dimensionless parameter given by

$$\eta = \alpha\theta/\lambda_0$$

When $\eta << 1$, $S(d)$ is proportional to the square of the Fourier transform of the intensity distribution of the source, as in the case of an experiment with visual stars. On the other hand, when $\eta >> 1$, the correlation function does not depend on the actual width of the source.[30]

The HBT experiment was carried out first with the two photomultipliers superimposed ($d = 0$), and then with the photomultipliers separated ($d = 1.8$ cm). It took six hours to run the experiment for each situation; the counting was done at five-minute intervals. Hanbury Brown and Twiss measured for each run the factor $S_e(0)/N_e$, where the experimental values of S and N come from the equations (2) and (3), that is:

$$\frac{m-1}{m}\int \alpha^2(\nu)n_0^2(\nu)dv \Big/ \int \alpha(\nu)n_0(\nu)dv, \tag{4}$$

which was obtained experimentally from the spectrum of the incident light, the direct current, and the gain and output noise of the photomultipliers. The HBT expression (2) thus provided the correlation between the numbers of photoelectrons detected at a time interval of observation. During the measurement, the photomultipliers C_1 and C_2 detected the outputs separately, and then the correlation between the intensity fluctuations was obtained through the correlator.[31]

As shown in Table 1, when the cathodes were superimposed, Hanbury Brown and Twiss observed a high correlation between the arrival times of

30. Ibid.
31. Ibid.

Table 1. COMPARISON BETWEEN THE THEORETICAL AND EXPERIMENTAL VALUES OF THE CORRELATION

	Cathodes superimposed ($d = 0$)		Cathodes separated ($d = 2a = 1{\cdot}8$ cm.)	
	Experimental ratio of correlation to r.m.s. deviation $S_e(0)/N_e$	Theoretical ratio of correlation to r.m.s. deviation $S(0)/N$	Experimental ratio of correlation to r.m.s. deviation $S_e(d)/N_e$	Theoretical ratio of correlation to r.m.s. deviation $S(d)/N$
1	+ 7·4	+ 8·4	− 0·4	$\gtrsim 0$
2	+ 6·6	+ 8·0	+ 0·5	$\gtrsim 0$
3	+ 7·6	+ 8·4	+ 1·7	$\gtrsim 0$
4	+ 4·2	+ 5·2	− 0·3	$\gtrsim 0$

Source: Hanbury Brown and Twiss, "Correlation between Photons" (ref. 27), 29.

photons in two coherent light beams in close agreement with the calculated theoretical value. However, the observed correlation decreased when the separation between the two photomultipliers was altered. When the photomultipliers C_1 and C_2 were separated by a distance d, the time of arrival photons at the photomultiplier C_2 was registered first, earlier than at the photomultiplier C_1 because the distance between the mirror and the photomultiplier C_1 was greater. As a result, *no* correlation between photons was observed.[32]

Comparing their theoretical values with the experimental results, Hanbury Brown and Twiss concluded that "the experiment shows beyond question that the photons in two coherent beams of light are correlated, and this correlation is preserved in the process of photoelectric emission." Noting a small difference between the theoretical and experimental results, they mentioned that it was probably due to defects in the optical system. Thus, the HBT experimental test seemed to confirm the possibility of constructing an optical interferometer using the same principles as the intensity interferometer for radio astronomy.[33]

Hanbury Brown and Twiss had therefore observed photons arriving at the same time at the two different photomultipliers. Yet their conclusion provoked a heated debate in the physics community, some of whose members claimed in articles and correspondence that the HBT experimental results were "nonsense." As Hanbury Brown and Twiss had used a low-intensity source, it would be expected that *only* individual photons were arriving at the half-silvered mirror in

32. Ibid., 29.
33. Ibid.

a given time interval, and hence *no* correlation between photons would be observed. In the following section, we present the controversial issues related to the HBT optical interferometer circa 1956–58.

THE CONTROVERSY

The first objection to the HBT experimental results arose in 1956 through an experiment conducted at the University of Western Ontario by Eric Brannen, of the department of physics, and his graduate student Harry I. S. Ferguson. Their motivation was to verify whether or not there would be a correlation between photons as found by the Hanbury Brown–Twiss optical interferometer. Commenting on the HBT results in the columns of *Nature*, Brannen and Ferguson openly claimed that "if such a correlation did exist, it would call for a major revision of some fundamental concepts in quantum mechanics," thus justifying the Brannen-Ferguson (BF) experiment. The BF experiment was virtually identical to the HBT experiment, aside from the detection process. Even before publishing their article, Brannen and Ferguson shared their results with Hanbury Brown and Twiss by suggesting that the HBT results could have been due to fluctuations in light intensity from the source and thus "it is possible that your radio telescope is reacting to a large number of photons, rather than to the behavior of individual photons." Hanbury Brown reacted to it, "I would welcome the publication of your experimental results, but I would reluctantly advise that you should not draw the conclusion that they disprove the correlation observed by us . . . I, personally, would welcome seeing a paper published which says that my own work is wrong . . . I am a great lover of scientific controversy, because I find I learn a lot from it."[34]

The BF apparatus used an electronic detection system to detect possible coincidences between two individual photons detected separately by the two photomultipliers over a period of time. Unlike in the HBT experiment, a linear photomultiplier was used and hence the intensity fluctuations in each detector were recorded, and a correlator combined the current outputs. Thus, the principal difference between the two detection systems was the fact that while Brannen and Ferguson detected individual photons at a time interval,

34. Eric Brannen and Harry I. S. Ferguson, "Question of Correlation between Photons in Coherent Light Rays," *Nature* 178 (1956): 481–82, on 482. E. Brannen and H. I. S. Ferguson to R. Hanbury Brown and R. Q. Twiss, 5 May 1956, and R. Hanbury Brown to E. Brannen, 5 Jun 1956, both in the Jodrell Bank Archives, University of Manchester, UK.

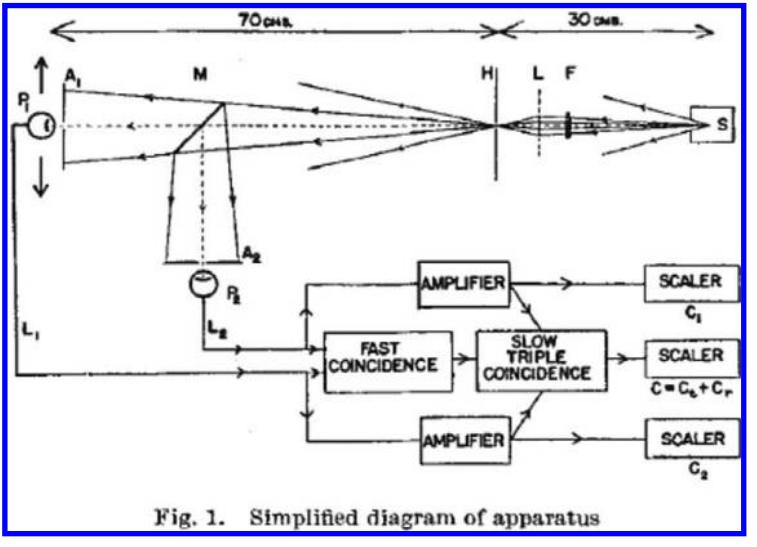


FIG. 4: The Brannen-Ferguson experimental diagram. *Source*: Brannen and Ferguson, "Question of Correlation" (ref. 34), 481.

Hanbury Brown and Twiss compared the intensity fluctuations of the outputs at the two detectors through a correlator. In the Brannen-Ferguson experiment (Fig. 4), as in the HBT setup, a light produced by high-pressure mercury passed through a filter and lens system, and then through a pinhole so that only monochromatic light reached the mirror; the beam was then split by a half-silvered mirror, and each separate beam of light was sent to the two photomultipliers; the coincidences were counted electronically.[35]

After performing their experiment, Brannen and Ferguson did not find any significant correlation (less than 0.01 percent) between photons in coherent light rays. Such an experimental result, according to them, agreed significantly with an experiment carried out *even before* the HBT experiment by the Hungarian physicist Lajos Jánossy (1912–78) and his research group from the Central Research Institute of Physics in Budapest. Brannen and Ferguson highlighted that Jánossy also agreed that if the existence of the HBT correlation between photons were confirmed, the foundations of the quantum theory should be revisited.[36]

35. Ibid., 481.
36. Ibid., 481–82.

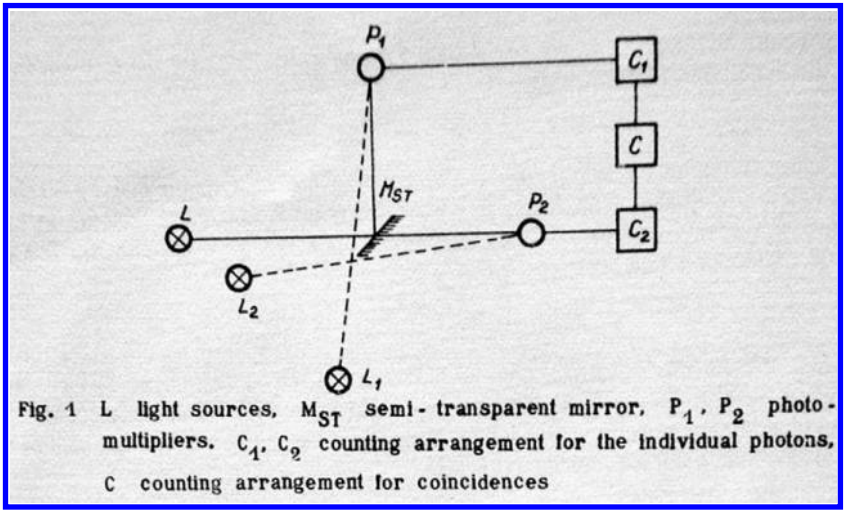


Fig. 1 L light sources, M_{ST} semi-transparent mirror, P_1, P_2 photo-multipliers. C_1, C_2 counting arrangement for the individual photons, C counting arrangement for coincidences

FIG. 5: The set-up of the AJV experiment published in 1955. *Source*: RHB, Box 18, Section E.64.

Jánossy, for his part, had always been interested in the foundations of the quantum theory, in addition to his interests in cosmic radiation, the theory of relativity, and the philosophy of physics.[37] By the early 1950s, Jánossy had become one of the critics of the Copenhagen interpretation of quantum theory.[38] This justified the motivation behind A. Ádám, Jánossy, and L. Varga (AJV)'s experiment "to investigate the validity of th[e] prediction of [the] quantum theory."[39] In their article, Ádám and colleagues proposed an experiment (Fig. 5) in which a low-intensity source L was split into two components by a half-silvered mirror and each component was sent to photomultipliers P_1 and P_2. The counters C_1 and C_2 were used to detect individual photons, and C recorded the coincidences. According to the authors and their interpretation of the prediction of the conventional quantum theory, if one assumes that photons are indivisible particles, they should be either in one component of a beam or in the other one after being split by the mirror. As a result, *no*

37. L. Pál, "L. Jánossy 1912–1978," *Acta Physica Academiae Scientiarum Hungaricae* 43, no. 1 (1978): I–IV.

38. See L. Janóssy, "The Physical Aspects of the Wave-Particle Problem," *Acta Physica Academiae Scientiarum Hungaricae* 1, no. 4 (1952): 423–67.

39. A. Ádám, L. Jánossy and P. Varga, "Observations on Coherent Light Beams by Means of Photomultipliers," translated into English from *Acta Physica Academiae Scientiarum Hungaricae* 4, no. 4 (1955): 301–15, in RHB, Box 18, Section E.64, on 57.

systematic coincidences between photons detected separately by the two photomultipliers would be observed.[40]

As a detection process, the AJV experiment used a coincidence counter to detect individual photons, as later used in the BF experiment, which was different from the HBT measurement process. In order to determine whether the coincidences detected by the photomultipliers were systematic—as HBT would later claim—or purely accidental—as traditional quantum theory seemed to predict—Ádám and his co-workers performed a version of their experiment with two independent sources, L_1 and L_2. As defined by them, the number of systematic coincidences would be obtained through the number of possible coincidences in the case of the source L (coherent light) *minus* the number of accidental coincidences from the independent sources L_1 and L_2 (incoherent light). Ádám and colleagues expected "merely" chance coincidences between the independent sources, and also systematic coincidences between the two components of the source L after splitting by the mirror.[41] After performing the experiment, they observed that the number of systematic coincidences between photons was approximately 0.6 percent—an experimentally insignificant figure—and concluded that "in agreement with the predictions of [the] quantum theory, the photons of two coherent light beams are independent of each other, or at least the biggest part of such photons are independent of each other."[42]

The AJV experiment seemed to confirm the principles of quantum theory as interpreted by Ádám and co-workers. However, a result that seemed to contradict quantum theory would appear a year later with the HBT experimental results. In their 1956 paper, Hanbury Brown and Twiss did not mention the AJV results. However, because they highlighted that a correlation between photons had never been observed, it seems likely that Hanbury Brown and Twiss could have had some knowledge of the AJV experiment. Thus, the HBT experimental results were significantly different from those of the BF and AJV experiments, and likewise in violation of the prediction of quantum theory, at least as discussed in the 1955 article by Ádám et al.

If the HBT experimental results were correct, it would be necessary to suppose that, for instance, photons—contrary to what Einstein had proposed in 1905—could be divisible particles, making it possible to detect them at the same

40. Ibid.
41. Ibid., 58.
42. Ibid., 63.

time at two different photomultipliers; or, as Hanbury Brown later put it himself, "one would have to imagine photons hanging out waiting for each other in space!"[43] Hanbury Brown continued, "[t]he basic trouble was that one can think about light in two different ways, as a wave or as particles. Richard and I had treated light as a wave which on arriving at the phototube causes the emission of a photoelectron . . . . However if you insist on thinking of light as a stream of independent particles like ping pong balls, which is what most physicists—especially particle physicists—prefer to do, then it is impossible to see how the arrival times of these particles can be correlated." This is why the HBT results were viewed by many physicists as "not only heretical . . . but patently absurd."[44] The standard concept of the photon thus seemed to be irreconcilable with the HBT correlation.

The criticisms provoked by the HBT results, as well highlighted by Hanbury Brown, revealed the way in which some physicists understood the concept of the photon at that time. Because of the low intensity of the source used in the HBT experiment, some physicists expected that only individual photons were arriving at a half-silvered mirror in a certain time interval. Hence, assuming the mainstream concept of the photon, as a billiard-ball model, each photon should be either reflected or transmitted by a mirror, and should only be recorded in one of the photomultipliers at a time. As a result, the chances of a correlation between photons should be theoretically zero—or extremely small, allowing for some defects in an actual experiment. Some physicists would not have believed that individual photons from a single beam could be detected simultaneously by two different detectors, as they had been in the HBT experiment. Of course, such difficulties disappeared completely if the incident light were considered to be a classical electromagnetic wave. In this case, there would be no doubt that two different points of an electromagnetic field might be correlated and detected simultaneously, even after having been split through a half-silvered mirror, since the incident light was coherent.

The first physicist to defend the HBT results was the American 1952 Nobel laureate Edward M. Purcell (1912–97) of Harvard University. At the time the HBT experiment was being carried out, Purcell had shown an interest in radio astronomy, working with his graduate student Harold I. Ewen on the

43. Hanbury Brown, *Boffin* (ref. 2), 121.
44. Ibid., 120–21.

construction of a horn antenna to investigate the interstellar medium.[45] In an effort to settle the dispute between the HBT and BF experimental results, Purcell wrote to the editors of *Nature*, attaching an article that would be his contribution to the discussions on the correlation between photons. In the correspondence, Purcell made it absolutely clear that he was serving as a "volunteer for the defense of [Hanbury] Brown and Twiss." Even though Purcell had thought that "if the issue is as simple as I believe it to be, it would be a pity to leave it unresolved for long," the controversy over the HBT experiment was just beginning. In fact, it would take approximately two years to resolve it.[46]

Purcell was the first to suggest that "the Brown-Twiss effect, far from requiring a revision of quantum mechanics, is an instructive illustration of its elementary principles," even though some physicists had criticized the HBT results based on it. In his interpretation of the HBT results, Purcell examined the problem through the statistical fluctuations of a system of bosons. Assuming that the probability of ejection of a photoelectron at a time T as a function of the square of an electric field (P) and an experimental constant (α) was $\ddot{y}\not{\phi}\S\alpha\bar{P}T$, Purcell determined the number of counts of the two photomultipliers at the same time interval separately, and then recombined the outputs from the two photomultipliers, finding a correlation in the number of photoelectrons detected given by

$$\overline{\Delta n_1 \Delta n_2} = \frac{1}{2}\overline{n_1^2}\tau_0/T \tag{5}$$

in which τ_o is a correlation time determined by the light spectrum, and T is a fixed time interval.[47]

Although the HBT equation and the Purcell equation were calculated using different theoretical approaches, Purcell's derivation represented, according to him, "the positive cross-correlation effect of [Hanbury] Brown and Twiss." The term "positive" seems to be related to the fact that Hanbury Brown and Twiss had observed a correlation between photons when they should not have, according to the traditional picture of the photon. The value of the cross-correlation could be different depending upon the nature of the particle used

45. John S. Ridgen, "Edward Mills Purcell, August 30, 1912–March 7, 1997," *Physics in Perspective* 13, no. 1 (2011): 91–103, on 96; Robert V. Pound, "Edward Mills Purcell 1912–1997," *Biographical Memoirs* 78 (2000): 3–24.

46. E. M. Purcell to the Editors of *Nature*, 9 Nov 1956, EMP, Box 3, Folder Calculations and Correspondence on Brown-Twiss Experiment and Brannen and Ferguson Correspondence, 1956–1957.

47. E. M. Purcell, "Question of Correlation between Photons in Coherent Light Rays," *Nature* 178 (1956): 1449–50.

in an HBT-type experiment. Using a beam of electrons, for instance, arriving at a nonpolarized mirror, the cross-correlation would be ($\overline{\Delta n_1 \Delta n_2} < 0$); or using a beam of classical particles, it would be ($\overline{\Delta n_1 \Delta n_2} = 0$); A null cross-correlation would only be found by sending classical particles to be split by a half-silvered mirror. These results should be expected, as stated by Purcell, since there might be a difference between the behavior of fermions, bosons, and classical particles.[48]

Unlike Brannen and Ferguson, Purcell suggested that "[t]he Brown-Twiss effect is thus, from a particle point of view, a characteristic quantum effect," being simply a consequence of a system of bosons. Such a quantum effect was a result of the "clumping" of the photons.[49] It seems that Purcell used the term "clumping" for the "bunching of photons," the probability of two photons reaching a certain point at the same time. In the HBT experiment, the light was produced by many different atoms, characteristic of a mercury source. Hence, when the first atom had already emitted a photon, the second atom began to be almost instantaneously excited and then emitted another photon in a short period of time. Thus, the use of the standard concept of the photon would no longer be controversial in the HBT experiment, since some photons might arrive in pairs at the half-silvered mirror. That was why the HBT results had showed a correlation between pairs of photon counts.

However, "[i]f one insists on representing a photon by wave-packets," as stated by Purcell, the HBT results could be explained as the probability of two trains, a stream of wavepackets in a random sequence, accidentally overlapping.[50] Therefore, Purcell suggested two ways to interpret the HBT results, depending on which picture of light was embraced: the wave aspect of light (in which the phenomenon could be explained through an overlapping wavepackets approach); or, the corpuscular aspect of light (according to which the phenomenon would be a signature of photon bunching).

Regarding the opposite experimental results, Purcell pointed out that the BF experiment did not detect, as in the HBT experiment, a correlation between photons because the observing time required to verify it depended on the resolving time of the apparatus and the stability of the source. That is, the HBT experiment was much more sensitive and accurate than the BF experiment.[51]

48. Ibid., 1450.
49. Ibid.
50. Ibid., 1449.
51. Ibid.

A copy of Purcell's correspondence and article was also sent to the protagonists of the debate. Aware of Purcell's work, Hanbury Brown wrote to him saying that he had strongly recommended the publication of his article to the editors of *Nature*. Recognizing the importance of having a Nobel laureate on his side in the controversy, Hanbury Brown also remarked that "although we can defend ourselves, it is nice to have an ally! [M]any physicists," as Hanbury Brown observed, "like to think of photons as independent little chaps who are loath to link hands as all proper bosons should."[52] Even after the development of Bose-Einstein statistics, from which the so-called bosons were born, some physicists still interpreted—in the case of the HBT experiment—a photon as a classical particle, a distinguishable entity described by Boltzmann statistics. Assuming the mainstream concept of the photon, the HBT results—an observation of a correlation between photons—did not make sense at all. Unlike some physicists, Purcell not only defended the HBT results, but also explained them using the properties of bosons. Hanbury Brown had agreed with Purcell that the HBT results were consistent with the elementary quantum theory, although he and Twiss had not used it in their theoretical approach published in 1956.

In correspondence with Purcell, Twiss highlighted that even though he and Hanbury Brown wanted to use the quantum theory in the first draft, they had "laid a great stress on the interpretation in terms of the corpuscular picture of light," using concepts such as the uncertainty principle and photon bunching. Nevertheless, the difficulty disappeared when the Belgian physicist Léon Rosenfeld (1904–74), who was one of the great defenders of the Copenhagen interpretation of the quantum theory[53] and who also worked at the University of Manchester, suggested a "sort of language at any price" to Hanbury Brown and Twiss. It was a "language" based on the semiclassical approach: the HBT experiment illustrated the wave aspect of light, and the quantum theory would only be used to interpret the detection process.[54]

52. R. H. Brown to E. M. Purcell, 15 Nov 1956, EMP (ref. 46).

53. On Rosenfeld, see Anja Jacobsen, *Léon Rosenfeld: Physics, Philosophy, and Politics in the Twentieth Century* (London: World Scientific Publishing, 2012); A. Jacobsen, "The Complementarity between the Collective and the Individual Rosenfeld and Cold War History of Science," *Minerva: A Review of Science, Learning and Policy* 46 (2008): 195–214; A. Jacobsen, "Léon Rosenfeld's Marxist Defense of Complementarity," *HSPS* 37 (2007): 3–34.

54. R. Q. Twiss to E. M. Purcell, n.d., EMP (ref. 46). Twiss, however, kept his inclinations for the bunching terminology, as he wrote to Rosenfeld, "I have also always referred to the bunching of photons as 'so called bunching,' though had I been in England I would have liked to

"In the end, after a deal of squawking, (which is not of much one at 12,000 miles range anyway)," as recounted by Twiss, who was then part of the Division of Radiophysics in Sydney,

> I was convinced that this was the better course since though it is certainly quite legitimate to me to use the photon concept throughout, as long as one knows exactly what [one was] mainly doing, it is only too likely to mislead the chaps who are always liable to forget that photons behave very different from classical particles . . . . However, nobody has any trouble believing that intensity fluctuations due to interference between waves emitted [from] different parts of the source can be correlated at different points in the field of the observer.[55]

The choice of an interpretation based on classical theory, instead of using a fully quantum theory, seems to have been pragmatic. On the one hand, even if Hanbury Brown and Twiss wanted to use quantum theory, there was doubt as to which concept of the photon should be taken into account. Thus, it was better to avoid an unclear interpretation of the phenomenon. On the other hand, a theoretical approach based on classical theory seemed to be much more understandable and acceptable because the phenomenon could be explained clearly through an interference effect. Nonetheless, the HBT interpretation came in for criticism.

Commenting on Purcell's interpretation of the HBT results, Brannen and Ferguson criticized Hanbury Brown and Twiss by claiming that "they expected [a] correlation even at low light intensities to the limit of only one photon being in the system at a time (to speak loosely),"[56] which would contradict the foundations of quantum theory. As described previously, as long as *only* individual photons were reaching a half-silvered mirror, it became extremely difficult to understand how Hanbury Brown and Twiss would have detected a correlation between two photons at two different detectors. Brannen and Ferguson claimed that they would carry out more experiments using a constant low-intensity source so that "only one photon will be in the system at a time, in order to keep away any effects due to photon bunching." According to them, the positive correlation observed in

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argue that this particular piece of the corpuscular terminology is not unhelpful." R. Twiss to L. Rosenfeld, 19 Oct 1956, Rosenfeld Papers, Niels Bohr Archive, Copenhagen.

55. Ibid.

56. E. Brannen and H. I. S. Ferguson to Edward M. Purcell, 29 Nov 1956, EMP (ref. 46).

the HBT experiment could be due to fluctuations in the source which could not be perfectly constant.[57]

Even though Purcell had mentioned the photon overlap model in response to the Brannen and Ferguson question about the HBT interpretation, he suggested that

> talking about *interference* of *photons* is the easiest way to go astray in such matters. To try to represent *a* photon by *a* wave-packet is asking for trouble. On the other hand the classical calculation, a la Brown and Twiss, of the fluctuations in P is a perfectly sound and rigorous procedure. The electromagnetic field *is* a classical field after all, which is why the Brown-Twiss effect only appears odd if one looks at it from a particle point of view; its oddness being simply the peculiarity of bosons.[58]

Clearly, Purcell did not like the idea of representing a photon as a wavepacket; however, the HBT theoretical model—based on the wave theory—seemed to be accurate and consistent with the experimental results. Because the electromagnetic field was a classical field, as mentioned by Purcell, one would expect to interpret the results through a semiclassical approach. Nonetheless, a difficulty arose from assuming the corpuscular picture of light. Such a difficulty might be solved by considering that the behavior of an ensemble of bosons differed from the behavior of classical particles. Of course, there would be no correlation between classical particles in a HBT-type experiment, but there would be systematic correlations between bosons because of the nature of those particles.

Hanbury Brown and Twiss, on the one hand, and Purcell, on the other hand, had different feelings about their involvement in the controversy. Hanbury Brown and Twiss, according to Hanbury Brown's later recollections, were almost excommunicated from the community of physicists.[59] Purcell wrote to Brannen cheerfully that "[w]e have had a lot of fun around here arguing about these questions, and I must say I have learned some physics in the course of it, which makes me grateful for the stimulation provided by the intrepid experimenters, yourselves included, who have gone back to really fundamental experiments."[60]

57. Ibid.
58. Ibid. (emphasis in original).
59. Hanbury Brown, *Boffin* (ref. 2), 121.
60. E. M. Purcell to E. Brannen, 17 Dec 1956, EMP (ref. 46).

In a draft article sent to Purcell that would eventually be published by Brannen, W. H. Wehlau, and Ferguson, the authors reiterated that the HBT correlation was not expected by quantum theory because "a single photon cannot be split!" According to Brannen and co-workers, Purcell had interpreted the HBT results as "an effect due to the interference between pairs of photons overlapping accidentally in time at the half silvered mirror." However, such an explanation could not be provided, according to them, because a low-intensity source had been used in the HBT experiment. Had Hanbury Brown and Twiss dealt with high-intensity light, photon overlap would have been plausible. "[S]ince one always has doubts about 'reasonable assumptions' when the elusive photon is concerned," as highlighted by Brannen and colleagues, a rigorous theoretical approach was desired. They therefore ended the draft with the following question: "If a photon is detected at one place how does this affect its capabilities of producing interference with another 'photon' at other places [?]."[61] It seemed to be far from straightforward to understand how the HBT results could be explained as interference between wavepackets overlapping when the experiment might deal with single photons. This was a paradox. The experiment conducted with low-intensity light—and therefore in the individual-photon regime—seemed to be explained successfully by a semiclassical approach, assuming the wave aspect of light. The HBT theoretical approach provided an explanation through a semiclassical model in a domain in which quantum theory should preside.

Answering Brannen's critics, Purcell claimed that "[p]ersonally I am not particularly fond of the explanation in terms of overlapping photons; it is both awkward to refine quantitatively and it is tricky unless one is very careful." Rather, he was merely giving an interpretation based on it in response to Brannen's question.[62] In fact, in his 1956 article he did not interpret the HBT results by means of the photon-overlap model, examining the problem statistically in terms of the number of photons arriving in a certain time interval. However, Purcell mentioned photon overlap as an alternative interpretation for the HBT results as long as one interpreted photons as wavepackets.

Unlike Purcell, the physicist Richard M. Sillitto, from the University of Edinburgh, did explain the HBT results through the overlapping wavepackets model, assuming that each separate wavepacket emitted by different atoms would be detected by a photocathode. Sillitto determined the mean square

61. E. Brannen to E. M. Purcell, 27 Feb 1957, EMP (ref. 46).
62. E. M. Purcell to E. Brannen, 1 Mar 1957, ibid.

fluctuations in the number of photoelectrons counted in a time interval. His result was similar to Purcell's. Because of an interference in the probability amplitude (caused by wavepackets superimposing coherently), Hanbury Brown and Twiss observed what Sillitto called "abnormal" fluctuations. That is, when overlapping wavepackets are superimposed coherently, there would be a higher probability of the emission of a pair of electrons within the overlap time. Regarding concerns about the concept of the photon, Sillitto claimed that his explanation did not imply that interference between photons, viewed as particles at that time, would create four or no photons. That is, if a photon could interfere with another one, in the end, there would have been either two more photons, or zero photons, which disagreed completely with the laws of conservation. Because "[t]he *photon is not a particle*," according to him, "it does not survive a counting process unchanged, and it is detectable only through its interaction with matter." Sillitto continued, "[w]hat does emerge from the argument above—and what can be understood in terms of this crude model and suffices to explain the experimental results—is that the interference between photons produces a distortion of the distribution time of the events by which photons are detected."[63] Once more the concept of the photon appeared as a kind of obstacle to interpreting the HBT results. Sillitto's approach, instead of assuming photons as classical particles, represented photons as wavepackets. As a result, "interference between photons" did not mean "interference between particles," but instead interference between the amplitudes of probability of overlapping wavepackets.

Another physicist who also participated in the discussions on the HBT results was Peter Fellgett from the Observatory at the University of Cambridge. Fellgett criticized the HBT "semiclassical assumption," according to which the probability of the emission of a photoelectron was proportional to the magnitude of the electric vector, claiming that such an assumption could be made successfully for radio astronomy, but not in optics. The HBT theoretical approach, as highlighted by Fellgett, "apparently conflicts with arguments of a thermodynamic nature." In order to show the limitation of the HBT semiclassical approach, Fellgett noted that a similar assumption would successfully describe the behavior of electrons, but not that of an assembly of bosons. He also argued that the electric field was not observable in optics, and that photons were not distinguishable particles (and therefore it was impossible

63. R. M. Sillitto, "Correlation between Events in Photon Detectors," *Nature* 179 (1957): 1127–28, on 1127.

to identify them between the source and the detector). Fellgett concluded: "semi-classical ideas, in fact, do not include the totality of our knowledge about the properties of radiation."[64]

Attempting to show the weakness of the HBT theory, Fellgett compared the HBT formula to another one, derived separately and previously by him and by the American physicist R. Clark Jones (from the foundations of thermodynamics) for the fluctuations in the number of photons absorbed by a body of emissivity ϵ in an enclosure. As a result, Fellgett concluded that the Fellgett and Clark Jones equations would rely upon the emissivity of the body, whereas the HBT one was a function of the quantum efficiency of the photomultiplier. This result thus seemed to show that the HBT theoretical approach was not in agreement with thermodynamics.[65]

Moreover, Fellgett desired "a refined experimental method" to observe a true correlation between photons since the HBT "experiment . . . belongs to the class in which an effect dependent on the 'wave' properties of light is observed in circumstances where the 'particle' properties predominate." The wave aspect of light would predominate, as mentioned by Fellgett, when there were many photons per unit volume. However, the particle aspect would come to the fore when a few photons occupied the same unit volume.[66]

Writing to Purcell after learning about Fellgett's article, Twiss suggested that he had begun to write an answer back, and "[t]o prepare this I had to plough through a vast number of papers on the fluctuations in radiation fields and came away with the firm conviction that the theory is in a pretty unhealthy mess. . . . I feel that much of the trouble is caused by trying to use thermodynamics in the wrong way."[67]

Owing to the theoretical controversy between the Fellgett and Clark Jones equations and HBT, Clark Jones, who worked for the Polaroid Corporation in the U.S., decided to circulate a report among physicists. Entitled "On the Disagreement between Hanbury-Brown and Twiss, and Fellgett and Jones," it discussed the principal disagreements between those derivations. Differing from Fellgett's point of view, Clark Jones wrote, "I believe that Hanbury-Brown and Twiss is correct in stating that our results are not applicable to a phototube, and their results are the correct one." By the time Fellgett and

64. P. Fellgett, "Question of Correlation between Photons in Coherent Beams of Light," *Nature* 179 (1957): 956–57, on 956.

65. Ibid., 957.

66. Ibid., 956.

67. R. Q. Twiss to E. M. Purcell, 30 May 1957, EMP (ref. 46).

Clark Jones had separately calculated the fluctuations in a body of emissivity based on principles of thermodynamics, their formulas agreed with each other. The fundamental difference between these derivations, as stated by Clark Jones, was that the HBT equation was calculated for a radiation source of finite temperature and photomultipliers at zero absolute temperature, while he and Fellgett assumed that the detector was in thermal equilibrium with the system. Taking into account these different experimental conditions, Clark Jones concluded that "both [equations are] correct in their respective field of application."[68]

The German-American physicist Leonard Mandel (1927–2001) also contributed significantly to the discussions on the HBT results, demonstrating that his theoretical approach to the problem agreed with the HBT and Purcell derivations, but disagreed with those of Fellgett and Clark Jones. However, Mandel's most significant achievement was to show that the number of photons arriving at a certain time interval obeyed the pure Bose-Einstein distribution when the coherence time (a period of time over which the light beams were still coherent) was much smaller than the bandwidth of light. Representing photons as Gaussian random waves, Mandel determined the correlation between fluctuations in two beams as a function of the degeneracy of the beams defined as the number of photons occupying the same Bose cell. His analysis was similar to the statistical approach of Purcell. Mandel also highlighted that "the degeneracy is also indicative of whether the wave or the particle properties of the beam predominate."[69] That is, if there were two or more photons occupying the same cell in phase space, the wave properties would predominate. Nonetheless, if there were only one photon in a single cell, the particle properties would be observed.

"Since the correlation depends essentially on two or more photons sharing cells in phase space," as stated by Mandel, "it depends on the degeneracy . . . [that] varies with the intensity of the beams." It seems that Mandel's equation provided the connection between the "bunching" of photons and the HBT correlations. Thus, if two or more photons shared the same Bose cell, there would be a higher probability of detecting two photons at the same time at two different detectors. However, there would be no correlation as long as

68. R. Clark Jones, "On the Disagreement between Hanbury-Brown and Twiss, and Fellgett and Jones," in RHB, Box 18, Section E.61.

69. L. Mandel, "Fluctuations of Photon Beams and Their Correlations," *Proceedings of the Physical Society of London* 72 (1958): 1037–48, on 1041.

only a single photon was occupying a cell. As a result, the HBT correlation between photons did make sense. Mandel concluded that "[t]he *correlation* is therefore appreciable only when the wave properties, as distinct from the particle properties, of the beam become evident. This confirms the view of Hanbury Brown and Twiss . . . that the effect should be regarded basically as a wave effect and shows it will be more difficult to detect in an experiment with light than with radio waves." The HBT results could be considered as a "wave effect" because the degree of coherence, which describes how correlated waves are, provided information about the phase of the beams through the correlation measurements.[70]

Other physicists who also participated in the HBT theoretical debate were the Czech-American physicist Emil Wolf (born 1922) and Lajos Jánossy. Wolf demonstrated theoretically that there was a correlation, whose value was proportional to the square of the coherence function, between two arbitrary points in a stationary optical field.[71] In another article, Wolf also determined, from a classical wave theory point of view, that it was possible to measure the degree of polarization of a light beam by using the HBT results.[72] Jánossy, whose work with Ádám and Varga we have already mentioned, discussed the problem from a classical viewpoint as well.[73] While Wolf had claimed that his theoretical results agreed with the HBT results, Jánossy mentioned that it would be necessary to perform more experiments to observe the "effect" since the Brannen-Ferguson experiment had not detected any significant correlation. Another physicist involved in the controversy was F. D. Kahn from the University of Manchester. Kahn's arguments will be discussed in the next section, as Hanbury Brown and Twiss made use of his arguments to defend themselves.

Because of the widespread criticism of the HBT experimental results, Hanbury Brown and Twiss decided to publish no more notes in the columns of *Nature* to argue in their favor, but a collection of four articles. The next section is dedicated to their more sophisticated experimental and theoretical arguments.[74]

70. Ibid., 1046.

71. E. Wolf, "Intensity Fluctuations in Stationary Optical Fields," *Philosophical Magazine* 2 (1956): 351–54.

72. E. Wolf, "Correlation between Photons in Partially Polarized Light Beams," *Proceedings of the Physical Society of London* 76 (1960): 424–26.

73. L. Jánossy, "On the Classical Fluctuations of a Beam of Light," *Nuovo Cimento* 6 (1957): 111–24.

74. These articles were published before the collection: R. Hanbury Brown and R. Twiss, "Correlation between Photons in 2 Coherent Beams of Light," *Nature* 177 (1956b): 1046–48; R. Hanbury Brown and R. Twiss, "Question of Correlation between Photons in Coherent Beams of

THE END OF THE CONTROVERSY

Responding to the criticism based on quantum theory, Hanbury Brown and Twiss revisited the most basic, yet controversial, concept from the Copenhagen interpretation of quantum theory: complementarity. Complementarity, first introduced in 1927 by the Danish physicist Niels Bohr (1885–1962) during the International Congress of Physics in Como, is the idea (in brief) that some concepts and pictures coming from classical physics, such as the wave and particle picture of light, are mutually exclusive concepts. For instance, if the wave aspect of light is observed in a specific experiment, the corpuscular one must be absent, even though those two concepts might be required for a complete description of a phenomenon. HBT's choice of complementarity to demarcate the frontiers between the wave and corpuscular aspects of light was drawn from their interaction with Rosenfeld, one of the most eloquent of Bohr's disciples.

Hanbury Brown and Twiss summarized Bohr's complementarity by writing that "a particular experiment can exemplify the wave or the particle aspect of light but not both." As the HBT experiment exhibited the wave aspect of light (if the light detected at each one detector was arranged to interfere, an interference pattern would be observed), the particle aspect would not come to the fore in the same experimental arrangement. Consequently, the concept of the photon was introduced only in the detection process, and not throughout the entire experiment. Accordingly, the wave or corpuscular aspect of light could be observed, depending on the experimental apparatus. If one used a linear multiplier to register the correlations between intensity fluctuations, the wave picture of light would be present. If, instead of using a linear multiplier, one used a coincidence counter able to detect individual events, the corpuscular picture would come to the fore.[75]

Light," *Nature* 178 (1956c): 1447–48; R. Twiss, A. G. Little, and R. Hanbury Brown, "Correlation between Photons, in Coherent Beams of Light, Detected by a Coincidence Counting Technique," *Nature* 180 (1957): 324–26; R. Twiss and R. Hanbury Brown, "Question of Correlation between Photons in a Coherent Light Rays," *Nature* 179 (1957): 1128–29.

75. R. Hanbury Brown and R. Twiss, "Interferometry of the Intensity Fluctuations in Light: I. Basic Theory—The Correlation Between Photons in Coherent Beams of Radiation," *Proceedings of the Royal Society of London Series A–Mathematical and Physical Sciences* 242 (1957): 300–24, on 300–01; R. Hanbury Brown and R. Twiss, "Interferometry of the Intensity Fluctuations in Light: II. An Experimental Test of the Theory for Partially Coherent Light," *Proceedings of the Royal Society of London Series A–Mathematical and Physical Sciences* 243 (1958): 291–319, on 291–92. Applications of the HBT interferometer to astronomy are in R. Hanbury

Although Hanbury Brown and Twiss had chosen "an alternative approach" based on classical wave theory to interpret their experimental results, the phenomenon could also be understood through the corpuscular aspect of light, considering the bunching of photons interpretation, or modeling it from a statistical quantum analysis. It was F. Kahn who explained the HBT results through another concept of the Copenhagen doctrine—the uncertainty principle—which was proposed in 1927 by the German physicist Werner Heisenberg (1901–76).[76] In Heisenberg's principle, a precise measurement of the position of a particle, for instance, causes indeterminacy in its momentum, and vice versa. Using the uncertainty principle to explain the HBT results, Kahn asserted that "[i]n a quantum picture it is the remaining uncertainty of the number of photons in each wave-train which makes the experiment work. The uncertainty is large in the radio experiment and small in the light experiment, with converse effects on the uncertainty in phase." That is, the uncertainty relation between the number of photons arriving at a receiver and the phase of the wave-train would be $\Delta n \Delta \phi \approx 1$.[77] In the HBT experiment, if the information about the phase were accurately obtained, there would be indeterminacy in the number of photons in the system.

After determining the rate of detection of photons for the HBT instrument, Kahn concluded that his theoretical model based on principles of quantum statistics agreed with the HBT semiclassical results and disagreed with the Fellgett results. Moreover, assuming that the electric fields at the two receivers were completely correlated, Kahn demonstrated that the probability of the coincidence of arrival of a pair of photons would depend on the visibility of the interference fringes as long as the light from those receivers were arranged to interfere in the HBT experiment.[78]

In an attempt to show the potential of the classical wave theory to explain the HBT experiment, Hanbury Brown and Twiss discussed a *Gedankenexperiment* described in Max Born's 1945 textbook *Atomic Physics*.[79] In a diffraction

Brown and R. Twiss, "Interferometry of the Intensity Fluctuations in Light: III. Applications to Astronomy," *Proceedings of the Royal Society of London Series A–Mathematical and Physical Sciences* 248 (1958): 199–221; R. Hanbury Brown and R. Twiss, "Interferometry of the Intensity Fluctuations in Light: IV. A Test of an Intensity Interferometer on Sirius A," *Proceedings of the Royal Society of London Series A–Mathematical and Physical Sciences* 248 (1958): 222–37.

76. Hanbury Brown and Twiss, "Fluctuations in Light: I. Basic Theory" (ref. 75), 302.

77. F. D. Kahn, "On Photon Coincidences and Hanbury Brown's Interferometer," *Optica Acta: International Journal of Optics* 5, nos. 3–4 (1958): 93–100, on 94.

78. Ibid., 95.

79. M. Born, *Atomic Physics* (London/Glasgow: Blackie & Son Limited, 1945).

grating experiment, one assumes that low-intensity light reaches a grating with two parallel slits, and that the light, which passes through the slits, will be observed on a screen. Although the source light is extremely weak, so that only individual photons reach the grating, "the experiment still illustrates the wave aspect of light, since the particle aspect can only really be brought out by observations in which the position of a single quantum is measured at two successive instants of time." Yet if such a position is measured in the experiment, the fringes of interference will disappear, and the corpuscular aspect of light would come to the fore. In other words, the appearance of the interference pattern would be a consequence of Heisenberg's uncertainty principle. Likewise, Hanbury Brown and Twiss argued that if one substituted the concepts of momentum and position of a photon from a diffraction experiment to those of energy and time, there would arise an "interference pattern in time, the beat phenomenon of the correlation experiment," because of the uncertainty in the energy of the photons. In the HBT experiment, the interference pattern could also be destroyed, as described by Hanbury Brown and Twiss, by using a highly monochromatic source of light or by inserting a prism in the apparatus. As a result, it would be possible to measure the energy of incident photons accurately.[80]

Hanbury Brown and Twiss ended by claiming that "since the radiation field can be treated classically in the case of the diffraction grating, it is only to be expected that it can be treated classically in analyzing the correlation experiment." However, if one insisted upon considering the particle aspect of light, the beat frequency (interference) in the HBT experiment would be "caused by the uncertainty in the energies of the individual photons which may be associated with either of the two Fourier components of the radiation field."[81] They continued by arguing that "[w]hen interpreting interference phenomena according to the corpuscular theory of radiation, it has been emphasized by Dirac (1947) that one must not talk of interference between two different photons, which never occurs, but rather of the interference of a photon with itself. This point was originally made for the case of spatial interference, as in an interferometer, but the arguments on which it is based are equally valid for temporal interference as in the phenomenon of a beat frequency."[82]

80. However, as we know from quantum optics developments after the HBT controversy, the disappearance of correlation would instead require sources producing one-photon states.

81. Hanbury Brown and Twiss, "Fluctuations in Light: I. Basic Theory" (ref. 75), 308.

82. Ibid.

In a more sophisticated theoretical model for the HBT experiment, Hanbury Brown and Twiss again used "a purely classical theory" to determine the mean square fluctuations in the emission current of a single phototube, and obtained an equation comprising two terms called the shot noise and the wave interaction noise. On the one hand, the shot noise term was interpreted as a consequence of the discrete nature of the electrons in the detection process, and hence it did not rely upon the quantization of the radiation field at all. The wave interaction noise, on the other hand, might be interpreted through the classical theory as due to the beats (interference effect) of the different Fourier components of the radiation field. However, as long as the corpuscular nature of light predominated, the wave interaction noise could be interpreted as the excess photon noise caused by the bunching of photons (as a consequence of Bose-Einstein statistics).[83]

Hanbury Brown and Twiss also improved their apparatus and carried out more experiments, again observing the HBT effect successfully.[84] Previously, they had compared "the theoretical performance of three equipments," considering the experimental parameters of each experiment, and found the time necessary to observe "a significant correlation" to be a thousand years for the Brannen and Ferguson experiment and 10^{11} years for the experiment of Ádám and coworkers, while the HBT experiment came off impressively at ten minutes.[85] Hanbury Brown and Twiss also applied their interferometer to optical astronomy, successfully obtaining the angular diameter of the bright star Sirius when compared to available measurements.[86] Such a successful application in astronomy helped consolidate recognition of the HBT effect in the field of optics.

Other evidence corroborating the HBT results was obtained by R. V. Pound and his student G. A. Rebka at the Lyman Laboratory of Physics at Harvard University in 1957. These researchers reported the validity of the HBT experimental results by observing the same correlation between photons.[87] Knowing

83. Ibid., 321.

84. Hanbury Brown and R. Twiss, "Fluctuations in Light: II. Experimental Test" (ref. 75), 291 and 308.

85. Hanbury Brown and Twiss, "Question of Correlation between Photons in Coherent Beams of Light" (ref. 74), 1448.

86. Hanbury Brown and Twiss, "Correlation between Photons in 2 Coherent Beams of Light" (ref. 74); Hanbury Brown and Twiss, "Question of Correlation between Photons in Coherent Beams of Light" (ref. 74).

87. G. A. Rebka and R. V. Pound, "Time-Correlated Photons," *Nature* 180 (1957): 1035–36; Rebka and Pound would perform an important experiment to test Einstein's general theory of relativity in 1959. See Kragh, *Quantum Generations* (ref. 1), 362–63.

previously through Purcell that Rebka and Pound would carry out a "local version of the experiment," Twiss expressed how a positive result could aid in the funding of the construction of HBT interferometers: "I hope that Rebka's experiment is soon successful and I should be very interested to hear how he is getting on. We pressed ahead with ours to buttress our claim for money to build our large mirrors and any independent check that all our results were not obtained by fudging would doubtless carry weight with the cash disbursers."[88]

In 1958, Brannen and colleagues actually observed a positive correlation between photons, thus confirming the HBT results as well and contradicting their earlier experimental results.[89] In private correspondence, Brannen recognized the HBT results as a physical phenomenon, noting that "[i]t seems to be well established now that our initial criticisms were unfounded. We [Brannen and Ferguson] thought (rashly it seems) that you were considering the 'splitting of individual photons', if you will pardon the phraseology, and our initial experiments were designed to test such a conjecture, which everyone would agree was contrary to quantum mechanics . . . . [A]s your calculations showed, our first experiments could not show such an effect whereas yours could."[90] The experimental controversy between the Brannen and Ferguson experiment and the HBT experiment thus ended in 1959. At the same time, the theoretical dispute between Fellgett and Hanbury Brown and Twiss also came to an end. Circulating a new report "The Resolution of the Controversy among Hanbury-Brown and Twiss, and Fellgett and Jones," Clark Jones confirmed the validity of the HBT fluctuations theory.[91] Moreover, Fellgett, Clark Jones, and Twiss also published an article, "Fluctuations in Photon Streams," in which they solved the theoretical controversy by displaying that "a number of objections to the thermodynamics approach originally put forward by Hanbury Brown and Twiss are invalid," and that both calculations were correct since they had been derived from different experimental conditions.[92]

88. E.M. Purcell to R. Q. Twiss, 10 May 1957, EMP (ref. 46).

89. E. Brannen, H. I. S. Ferguson, and W. Wehlau, "Photon Correlation in Coherent Light Beams," *Canadian Journal Physics* 36 (1958): 871–74.

90. E. Brannen to R. Hanbury Brown, 22 May 1959, RHB, Box 18, Section E.61.

91. R. Clark Jones, "The Resolution of the Controversy among Hanbury-Brown and Twiss, and Fellgett and Jones," RBH, Box 18, Section E.61.

92. P. Fellgett, R. Clark Jones, and R. Twiss, "Fluctuations in Photon Streams," *Nature* 184 (1959): 967–69, on 968.

Reflecting on the controversy in 1991, Hanbury Brown noted that "we had quite a hard job persuading people that to talk about the behavior of a beam of light as though it is a stream of independent photons which preserve their individual identities from emission to absorption is a gross misuse of the concept of a photon and gives the wrong answer." Rather, "[w]e had to persuade our opponents, many of whom were surprisingly irate, that there is no satisfactory mental picture of light which gives the right answer to this particular problem and that the only way of getting the right answer was to do mathematics." Although Hanbury Brown and Twiss had had to face the severe criticism and some funding issues to support the construction of the stellar interferometer, "[a]ll this controversy taught many physicists something new about the nature of light."[93] We can note that after solving the debate surrounding the HBT laboratory experiment, Hanbury Brown and Twiss finally constructed their Narrabri Stellar Interferometer located 370 miles north of Sydney in Australia.

SOME CONCLUDING REMARKS

The controversy surrounding the HBT results contributed to a debate on the concept of the photon in the late 1950s, mobilizing twelve theoretical and experimental physicists around the world from different institutions and backgrounds. It was a controversy that taught many physicists about the boundary between theoretical and experimental physics, and in particular, about the nature of light. The HBT controversy arose in the community of physics as a result of the way some physicists understood the concept of the photon from old quantum theory. As highlighted by Richard Sillitto, "[i]t is one of the interesting features of [the HBT] result that it cannot be understood in terms of the crude—too crude!—model of a beam of light as a stream of discrete, indivisible, corpuscular photon."[94] Thus, the HBT results seemed to contradict the predictions of old quantum theory, which assumed a corpuscular picture of the photon. The HBT debate revealed how some physicists still viewed photons as classical particles, even after the development of the Bose-Einstein statistics. As discussed previously, the HBT experiment was explained through two different pictures: the wave aspect of light (according to which the

93. Hanbury Brown, *Boffin* (ref. 2), 121–23.

94. R. M. Sillitto, "Light Waves, Radio Waves, and Photons," *The Institute of Physics—Bulletin* 11, no. 5 (1960): 129–34, on 131.

radiation is a classical electromagnetic wave and the matter is quantized); or, the corpuscular aspect of light (in which the HBT results are explained through "photon bunching" as a consequence of the behavior of the bosons).

Nowadays, the HBT correlation is known as the HBT effect (as it was called initially by Purcell in 1956): any correlation between intensity fluctuations in two photomultipliers when using a thermal, or chaotic, source. With the benefit of hindsight, it is easy to understand why the HBT experiment was able to detect such a correlation: HBT's work did not deal with single photons because of the characteristic of the source used in their experiment. An anti-correlation between photons (or "photon anti-bunching") would be detected in the 1970s after the development of the laser and quantum optics. The HBT effect became such a landmark in physics that physicists and textbooks came to describe it as heralding the birth of quantum optics. Even when Hanbury Brown and Twiss were awarded the Albert Michelson Medal in 1982, their contribution to physics was summarized with the words: "it is safe to say that much of the early history of quantum optics has its roots in the Hanbury Brown–Twiss effect and that this phenomenon can rightly be viewed as a cornerstone of modern optical science."[95]

In fact, as asserted by the American physicist Roy Glauber, who was awarded the Nobel Prize in 2005 for the development of the coherent states of the electromagnetic field, the HBT effect indirectly inspired his own work. In his autobiography, Glauber remarked:

> [T]he late 50's proved to be an exciting time for many reasons. A radically new light source, the laser, was being developed and there were questions in the air regarding the quantum structure of its output. That was particularly so in view of the surprising discovery of quantum correlations in ordinary light by Hanbury Brown and Twiss. . . . That was the period in which I began to work on quantum optics with a surmise that the Hanbury Brown-Twiss correlation would be found absent from a stable laser beam, and then followed it with a sequence of more general papers on photon statistics and the meaning of coherence.[96]

95. Hanbury Brown, *Boffin* (ref. 2), 120.

96. R. Glauber, "Autobiography," Nobelprize.org, at http://www.nobelprize.org/nobel_prizes/physics/laureates/2005/glauber-autobio.html (accessed 8 Aug 2012). Glauber's quantum explanation of the HBT results will be examined in a following paper, together with the concept of the photon that emerged with quantum optics.

In his 1963 article, Glauber explained the HBT results by the quantization of the electromagnetic field in optics.[97] A question arises: Which concept of the photon emerged from quantum optics? Was such a discussion on the agenda of physicists at the time? If not, why not? Were physicists pragmatists in the sense of using only mathematical arguments, instead of expending philosophical energy in an effort to understand or interpret what the photon was? These questions merit further investigation.

The HBT episode illustrates how long it took for the concept of the photon to coalesce to what we take for granted today—unsettled in 1905, 1916, and 1927; unsettled even in the 1950s, fifty years after its introduction. The HBT experiment showed that physicists could not interpret photons by means of the "too crude corpuscular model" when a thermal source was used. Using laser and certain quantum states of light as sources, however, physicists actually observed in the 1970s and 1980s an anti-correlation between photons. Photons, now mathematically represented by number states, came back to physics. The presentation and discussion of such experiments, however, is a matter for future studies.[98]

The success of the semiclassical approach also contributed to physicists' not considering the photon concept, or the quantization of radiation, before the detection process. The HBT argument, for instance, was sufficient to explain

97. While it is beyond the scope of this paper to analyze Glauber's work, his use of quantum field methods to deal with light coherence should be noted (Brown and Pike, "Optical and Optoelectronic Physics" [ref. 5], on 1439–41). According to Bromberg, Glauber "had worked in quantum field theory and nuclear physics." Even though at the time he "had not himself being doing research on optical coherence, . . . he had been in touch with it because some of the theoretical and experimental work was being done in the Harvard physics department." And yet, he asked himself, "how can Hanbury Brown and Twiss's results, and the completely coherent character of laser light, be embodied in a fully quantum-mechanical theory?" In Joan Bromberg, *The Laser in America: 1950–1970* (Cambridge, MA: MIT Press, 1991), on 109. In contradistinction, the physicists involved in the debate on the HBT experiment did not mobilize such resources. It deserves to be investigated how much of this choice was related to their training and how much was due to the fact that they explained HBT results without looking for a general treatment of light from which to derive the HBT results.

98. John F. Clauser, "Experimental Distinction Between the Quantum and Classical Field Theoretical Predictions for the Photoelectric Effect," *Physical Review* 9 (1974): 853–60; H. J. Kimble, M. Dagenais, and L. Mandel, "Photon Antibunching in Resonance Fluorescence," *Physical Review Letters* 39 (1977): 691–95; C. K. Hong and Leonard Mandel, "Experimental Realization of a Localized One-photon State," *Physical Review Letters* 56 (1986): 58–60; Philippe Grangier, G. Roger, and Alain Aspect, "Experimental Evidence for a Photon Anticorrelation Effect on a Beam Splitter: A New Light on Single-Photon Interference," *Europhysics Letters* 1, no. 4 (1986): 173–79.

the experimental results. Moreover, looking back to the early development of quantum theory, semiclassical models were used separately by Erwin Schrödinger to examine the Compton effect and by Guido Beck and by Gregory Wentzel to discuss the photoelectric effect.[99] Even in the late 1960s, the physicists Willis Lamb and Marlan Scully revisited the semiclassical approach and analyzed the "Photoelectron Effect without Photons."[100] Nevertheless, it seemed that it was only with the development of a full quantum theory of light and later experiments that a sophisticated concept of the photon came to the fore, and hence semiclassical models no longer provided a completely satisfactory description of light.

Although the concept of the photon seemed to have been fixed after the early development of quantum theory, the HBT debate helped physicists to revisit and reinterpret, or understand, the concept of the photon. The HBT experiment also illustrates how the performance of experiments has contributed to interpreting the foundations of quantum theory.

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99. See E. Schrödinger, "Über den Comptoneffect," *Annalen der Physik* 82, no. 2 (1927): 257–64; G. Beck, "Zur Theorie des Photoeffekts," *Zeitschrift für Physik* 41, no. 10 (1927): 443–52; G. Wentzel, "Zur Theorie des photoelektrischen Effekts," *Zeitschrift für Physik* 40, no. 8 (1927): 574–89. Those approaches were revisited in the 1980s by J. Strnad, "The Compton Effect: Schrödinger's Treatment," *European Journal of Physics* 7 (1986): 217–21; and J. N. Dodd, "Compton Effect: A Classical Treatment," *European Journal of Physics* 4 (1983): 205–11.

100. W. E. Lamb and M. O. Scully, "The Photoelectric Effect without Photons," *Polarisation Matière et Rayonnement,* Jubilee Volume in Honor of Alfred Kastler (Paris: Presses Universitaires de France, 1969), 363–69.